

Mark Solms ~ Active Inference Insights 017 ~ Affect, Consciousness, Dreams

Insights | Active Inference Insights 017 ~ Affect, Consciousness, Dreams | 1:23:30

hello everybody and welcome back to
active inference insights today I am
honored to be speaking with Professor
Mark Solms Professor Solms is a
neuropsychologist and psychoanalyst best
known for his discovery of the full
brain mechanisms underlying dreams and
his integration of psychoanalytic
theories with modern Neuroscience more
recently he's been focusing on active
inference and what it might bring to
bear on the hardest problem of them all
Consciousness mark thank you so much for
joining me it's a yeah as I said in the
introduction it's a real honor and
privilege it's it's great to be here
Darius thanks very much so yeah I mean
you have been in the field of cognitive
Neuroscience for many decades but I
guess have only recently in the last 10
years or so have really taken an
interest in in active inference and
seeing you know as many people have I
think when I read your book um The
Hidden Spring which is a wonderful read
and I think what I really liked about it
I've been looking I'm very interested in
the hard problem and trying to kind of
see different perspectives with which we
can take to to approach the hard problem
and I've been looking for what I've been
calling the synthetic a priori of
Consciousness and what I mean by that is
can we have a can we determine a
function of cognition that needs to be

conscious a priori so for example you could say attention well I can imagine offline attention I'm just sort of pinging around without it being online and I thought you got very close and I I will not say determin you know I will not say in a determined fashion because David Chalmers will shs at me but I think you got very close with this notion that affect is a good candidate for what I would call the synthetic a priori of Consciousness now affect is can be seen as a kind of loose term so I think where we might want to start is just if you could give a kind of brief definition of what you mean by affect because obviously um maybe what you and yach panp in basic emotion Theory think is a might differ from a constructionist account of aect and so on so maybe if we could just start laying down what affect is before we get to

Consciousness thank you um yes that's a good uh point to start with because as you say different people do mean different things by that term um so what I mean by affect is first and not not foremost but first that it is valanced in other words there's a goodness and Badness um to affect

and um appropo of your opening remarks about a um aiori synthetic function uh the I think I should mention that the goodness and Badness of affect um is it applies only to the subject of the affect so it's good from my point of view or bad from my point of view it doesn't mean that it is objectively or

universally good or bad so an affective state uh is a valanced state it has goodness and Badness the second um and perhaps this is the uh when I said I don't want to say first and foremost veilance because perhaps this is the foremost characteristic is that it is qualified by which I mean uh it has a specific uh differentiated quality there are there are different affects so although all affects share the the um property of veilance of goodness or Badness um each affect has its own unique quality um I don't mean to say that they aren't intermediate forms of affect but what I mean is that the that the feeling of hunger uh is qualitatively different from the feeling of thirst is qualitatively different from the feeling of sleepiness and so on and uh this is important um because you you you you can't you can't say um I have uh currently six out of 10 of sleepiness and six out of 10 of thirst therefore um I have uh 12 out of 20 of total bily need and all I need to be is sleep uh if you if you did that you would reduce the the the the common uh denominator uh the the the continuous variable of need but you would die uh that is because uh sleepiness and thirst are not the same thing they are qualitatively different they are categorical variables so so that's for me uh the essential features of affect that they valanced um which which implies they're subjective their

veillance is subjective uh and they are
categorical uh qualitatively
differentiated uh

States okay that's really useful so I
think before we get into the qualitative
distinctions because I you know uh
people may have seen your debates with
Lisa Felman Barrett um and how deep that
distinction runs was kind of a sticking
point in your in your conversations
before we go there we don't even
necessarily need to go there I think the
really interesting point there is that
good or Badness is only good or Badness
for a subject I've been thinking about
this in terms of relevance or
prioritization that whatever enters into
the space of Consciousness is always
prioritized or relevant to the
uh to the agent now I think there's a
second question that I will come to
about are we not putting the C for the
horse by H inferring the presence of an
agent for whom something is good or bad
but before I go there folding in
something like salience relevance or
priority is that uh built into the
goodness or Badness of the signal of the
phenomenal state or is it or does it
necessarily underly or underwrite that
phenomenal

State um the way I look on it is that
the the underlying category is a
category of need uh you do not feel all
your needs so for example your need for
energy resources um is met by your
constantly burning up the uh glucose uh
deposited in your ad

tissues but you don't feel anything um
when that um need is prioritized you
then feel hunger um so the
prioritization and the becoming
conscious of the need are I think
intimately bound up with each
other and so what therefore does that
say for this kind of synthetic aior of
Consciousness in so far as if relevance
or salience or priority whatever you
want to say
is it it's difficult to imagine being
conscious of something which is not in
some way Salient to
you does that put the that notion of
salience uh which we could you know I've
been proposing that Consciousness might
be something like a relevance filter or
a salience filter would that make that
function that relevance function more
akin to this synthetic aior of
Consciousness than affect per se
I think that uh the answer to that is
bound up with uh what do we mean when we
say we we are prioritizing one need over
another it's not it's not the
prioritization is not an end in itself
it's a prioritization for Action uh you
can't you can't do everything uh at once
um in terms of um this prioritization uh
uh uh imperative and I'll explain what I
mean by that uh and has everything to do
with when you say an aoi synthetic
function um it has everything to do with
what function we are talking about so
what I'm saying is that that function is
not merely the function of need or the
registration of a need uh it's the

feeling of the need and what functionality is added by the feeling of the need has everything to do with the action bottleneck that I was talking about the prioritized need is felt so that you can monitor how well or badly you're doing uh in your attempts to meet that need the nonprioritized needs are relegated to automaticity the actions that you perform in relation to the nonprioritized needs do not have to be felt because uh they are being met or at least attempts are being made to meet them outside of Consciousness by existing uh fixed um policies the prioritized need which is felt uh the the function of the feeling is precisely because you're in a state of uncertainty um the the the automaticity uh doesn't cut it automaticity is dangerous uh because you don't know what to do so you're feeling your way through a novel situation or through an unpredicted situation and um the the the feeling tells you whether what you're currently doing the policy you're currently following whether it's working or not so it underwrites Choice uh it underwrites the possibility uh of changing your mind on the Fly um and then of course learning from the experience which which which results in the laying down of new predictions which may in future uh be executed unconsciously it's while you it's sort of predictive work in progress I suppose is a is a good way um of describing it and I want to um make explicit what I in something in what

I've just said when I said that this underwrites Choice um it means that it underwrites voluntary action so this is the mechanistic difference in my view between voluntary and automatic or autonomic action so yeah okay that's that's interesting there are some things that I think we can fold in from an active inference perspective because um you were talking you were talking about almost metacognitive monitoring of how well you're doing at minimizing free energy and that actually has been fleshed out in in active inference as a COR jofy and Corelli paper all the way back in 2013 but recently people have been talking about um era Dynamics which is this awareness of how well you're doing at minimizing free energy over a span of time and they have Associated that with veilance my question here is more therefore philosophical so it's not really pulling that in aart um although there might be some people who who would encourage me to do do so but rather why couldn't that uh monitoring be done offline so what is it about and this really gets to the heart of this notion of a synthetic priority of Consciousness I can imagine in some sense a robot in some fault experiment who was able to offline attribute goodness or Badness to their experience and use that in a functional manner to guide their expected free energy calculations let's say what is there something critical to

experience per se which makes it
adaptive um for these free energy
minimizing systems and their
metacognitive capacities to track their
ability to minimize free energy yes I
believe that there is before I um
explain why I just want to flag maybe
for later discussion so I don't I don't
divert us from the question that you're
the line of questioning that you're
currently following that I'm not so sure
that I would describe the mechanism that
um I just articulated as a
metacognitive uh mechanism I I think
there's a difference between um the
feeling um and what the feeling is about
in other words there's a difference
between uh the the the raw feeling which
is the registering uh of what's
happening to your in in the terms that
you were just using uh to your free
energy and um you're cognizing
uh representing to use another language
um what uh in in the context within
which I'm feeling this what is causing
me to feel this so sure I yeah I
certainly I certainly agree with that
yeah I didn't mean to say in terms of
metacognitive purely this notion that it
can be done in some pre-reflexive
non-conceptual non-representational
manner purely this notion that there is
at least embedded in phenomenal space
some self referential but it could be
very low level like lived experience I
am doing well I am doing badly but it
doesn't have to be a higher order
narrative um I doing well I'm doing

badly okay good now um to your question
you ask why can't this function that
I've just described uh as the mechanism
of feeling uh you're asking why can't it
go on in the dark uh why why why is it
why is it necessary for it to be felt so
let me if you'll just bear with me for a
for a couple of minutes let me unpack
the mechanism in in slightly more
pedantic detail um we're talking about a
self-organizing system uh which is which
is intrinsically imbued uh with the
um the the um function that it must
continue what it does uh must uh
maximize its chances of continuing to
exist so that's a sort of pre that's a
starting point of of of any complex
dynamical self-organizing system so
there is a kind of intrinsic I don't say
kind of there is an intrinsic
intentionality in such a system it's
trying to achieve something I don't mean
it's metacognitively trying to achieve
something I just mean it has it has a
goal
directedness let me let me put it that
way um now
it's I then said that um the monitoring
of how well or badly I'm doing uh this
this can this veilance function um
that's what veilance is because it's
just to the extent that what I'm doing
um is moving me out of my viable bounds
is making me less likely to survive to
and vice versa you know when it's moving
me back towards my viable bonds that
just is good or bad for the system
the but again I need to emphasize the

point that I emphasized earlier it just is good or bad for the system in in in uh Nagel's sense it's it's it's good for the system and good to the system you know so it's not it's not good or bad in any objective sense it's good or bad only in a subjective sense now you're saying why can't that sort of monitoring uh be done um in the dark I'm saying it can that's why I made the point earlier about the difference between the continuous variable of total need or total free energy um and the categorical variable um of which which need am I talking about here categorical variables just are distinguished qualitatively as I said earlier thirst and sleepiness just are not the same thing uh so they are categorical variables uh that means they're qualitatively different so what I've just described is something which mechanistically is a felt state I'm saying I'm saying from the point of view of the system there is an an existential uh variable uh called veilance in other words this this is uh good or bad for me uh for my continued existence whether or not this is metacognitively reflected it doesn't matter there is this thing which just is good or bad and just is subjective and it just is qualitatively differentiated so if you're saying that there's a certain function U that the system has uh which is intrinsically subjective intrinsically valanced um and intrinsically qualitatively distinguished I'm saying that is a

feeling so if you say say how can that
go on in the dark I would say what do
you mean

uh this intrinsically is something that
is felt that's the me a mechanistic
description of what a feeling is and I
think that the crucial step um that uh
needs to be um uh identified there um is
that is it justifiable to take the point
of view of the system in other words
what is it that justifies one uh to take
the Viewpoint of the system and I hope
you can see in what I

said the in the intrinsically subjective
nature of the things that I was talking
about there that justifies us the self
of the system justifies Us in taking the
subjective point of view of the system
and if you do that from a subjective
point of view the mechanism I've just
described um just is a feeling that's
what we mean by a feeling so um I I I I
hope I've been plain

clear enough you have well you have been
for me I mean I'm familiar with these
arguments of sort of intrinsic
intentionality I mean as you know they
go all the way back to haiger and this
notion of taking a stance on your own
being and the idea that it's kind of
open-ended we are naturally as you say
directed towards this kind of open-ended
Futurity and that's gets fleshed out in
you know I've recently just finished
Evan Thompson's mind in life and he is
wonderfully clear on that about this
idea of kind of we make we the unel
comes out of our own stance that we take

upon the world I guess I still have a
kind of residual Charmers
skepticism which is does and I think
therefore it's necessary to get kind of
clear on the axioms that you're taking
here does taking an intentional stance
on your own being require Consciousness
maybe intent well intentional is
actually not necessarily the wrong word
here but I mean can you have a
self-organizing complex dynamical system
and I think there's an interesting point
here about the imperative but I'll come
back to that but can you have a complex
self organizing system which is
autopoietic in the sense it is
intended towards its own um self
persistence which need not be conscious
or at least conscious in the way that we
think of a conscious as subjectively
conscious I can almost imagine that you
have a nonsubjective
Consciousness I.E an objective
Consciousness if one could imagine and
such a thing which might in some
ways uh suit that self-organizing
process because it gets rid of bias and
uh overfitting your data and
confirmation bias and all this other
stuff I recognize we've gone slightly
off the beat and track here but I'm just
wondering whether there's like one thing
that I'm missing maybe it's an axiom or
maybe it's an argument I'm not sure
about exactly why autopoietic self
organization entails subjectivity per
se um well well let me start um with an
appeal to Authority which is not a good

thing uh and then uh let me add a um semantic point which is not a good thing and then I'll I'll answer your question properly sure the appeal to Authority I have to say um I was very pleased that Tom Nagel wrote a commentary on my book in the journal yeah he wrote it in the Journal of Consciousness studies um and there he he said in in in relation to exactly the question you've just asked that he thinks that it's it's a non nonstarter the idea of an affective zombie so he accepts that it cannot be an affective zombie this this is a this is a nonsecretor um but that's an appeal to Authority I'm just saying no less a person the nature of subjectivity uh he he says he he accepts that there cannot be an an affective zombie in other words there cannot be a system equipped with the functionality that I've just described that is a zombie because if it were a zombie it wouldn't have the functionality I've described but that's just an appeal to Authority then there's the semantic Point um I want to make clear that I'm talking about feeling okay and you asked me earlier about affect and you said different people use the word in different ways and they mean different things by it um I Hope from my description of how what I mean by the word you can see why I use it synonymously with feeling um I like to use the word feeling when I'm discussing the issues that you're raising because

it just is an oxymoron to speak of a feeling that you don't feel uh there's it's not a feeling if you don't feel it uh and and so the thing denoted by that word um is the thing I'm talking about now I said already that I accept that that's just a semantic point and you know but it it leads to something more than a semantic point when I say the thing I'm talking about is the thing denoted by the word feeling therefore the thing I'm talking about cannot be unfelt because uh that's the thing I'm talking about the thing I'm talking about is the thing that we call feeling now that leads me to the substantive answer to your question I really do believe that this this question as you formulated it um I I I recognize of course as you formulated it derived from or with um with reference to David Chalmers's formulation of it um I don't think this question would ever have arisen if it wasn't for the fact that uh we were using as at the time that Chalmers wrote his famous paper and the book that followed it um and formulated this term the hard problem of Consciousness um we were using as our model example of Consciousness visual Consciousness uh um This was um perhaps uh inevitable because Consciousness is dominated by Vision uh and uh it was perhaps um because Francis Crick used it as his model example in his book uh the the um uh what was it called the astonishing hypothesis which came out

one year before uh charm's paper um the thing is that I think that was the wrong place to start that all of those questions arise only because vision is not an intrinsically conscious function uh even the highest of visual um functionality uh in other words uniquely human cortical um visual functions like reading with comprehension can be done without Consciousness so the question why doesn't all this information processing go on in the dark without any inner field which is to quote exactly the the phraseology that Charmers used I think that that's a legitimate question and I can see why the puzzlement arises because you can do exactly the same thing without Consciousness so you know like reading with comprehension why on Earth do you ever why does there have to be something it is like to read if the exact same thing can done without there being something it's like to do it now you then once you've got this puzzlement then somebody like you a reader of Charmers comes to my argument and says well why is it inconceivable that this thing couldn't be done by an affective zombie and I'm saying well that's a silly question because look at what I've just told you you know I've told you what the function is I've told you that it is intrinsically subjective it's intrinsically existential it's intrinsically veence uh and it's qualitatively distinguished and if you describe such a mechanism which is necessarily

registered by the system then you're describing a feeling so what do you mean it could be not felt it wouldn't be a feeling if it was not felt and I've told you what the feeling does you know and yeah I guess I guess I need to be a little bit more philosophically robust and it's come to mind here that one can take charma's claims and and that 95 paper and the book in multiple ways so one is why um couldn't this functionality be done in the dark and I take and I agree actually that feeling a prior in and of itself by definition has to be felt so that's something that you can have which needs to be felt so that that notion of in the darkness doesn't really stand when it comes to affect the another the other part of David charma's argument is how do neural processes give rise to subjective states to what it feels like to be me or you and I guess that is a different direction which I presume you probably don't think there is an answer to and I guess what that leads me back to is when I'm talking about the synthetic a prior of Consciousness sure feeling gets us there I guess what I was saying is if we go if we have the sort of view from nowhere we can go back all the way to the evolutionary start and let's say we did start as non-conscious beings why was it functionally adaptive for human beings to develop feelings in the first place right if this metacognitive non-representational uh self-referential

process what it became imbued with
feeling not I'm not going to ask how
that happened because no one knows but I
guess the question here is why did that
happen and that's when I say well
couldn't couldn't have happened in the
dark sure feelings themselves can't
happen in the dark per se but the
function that feelings fulfill namely
this monitoring process why why did that
have to evolve so as to contain qualia
subjectivity and and so on that's the
kind of Crux of the question I
guess um I think you're I think either
you're asking three questions or I've
got three answers uh to your question so
I'm so I'm not so okay so I don't okay I
think there's actually only one here one
okay so I'm not fussed about how I'm
gonna lose my train of thought I'm an
old go go go go go go please me go go go
so um if I haven't already lost it um ah
yeah what I want to say first of all is
the um Mary uh the visual
neuroscientist um if if she were an
affective neuroscientist uh if she were
to describe the to know everything there
is to know about the mechanism of
feeling it would necessarily predict
that it feels like something and and she
would know something about what it is
like to have a feeling from the
mechanistic account of it um that's
that's that's the the first thing that
you were saying in your question brought
that answer to my mind but I think you
said you concede that point uh I'm not
uninterested I'm uninterested I well now

I'm interested but I was uninterested in the how and I'm and I'll give you the feelings need to be felt I guess my question is evolutionarily why does the function of feeling need to be felt yes so um that leads to or or that to get to that I've got to go via a second thing that I I heard you saying which is um you know how does the physiological process cause the subjective feeling State and there I have to make clear that I am a dual aspect monist um I'm you may want to explain what that is for the audience yes I I I will um I'll put it uh very simply uh when I wake up in the morning uh before I open my eyes I experience my existence I think to myself I am Mark Psalms thank God I'm still alive that's my mind okay then I stagger over to the bathroom and I look in the mirror um and I see this body uh uh called Mark alms and I say uh there is further evidence that I still exist thank God uh now please note both of those things I'm calling them Mark salms u in other words these are two um two phenomenal uh two appearances two observational perspectives upon one and the same thing so there is the subjective being of Mark PMS that I experience mentally when I wake up before I open my eyes and then there is the objective bodily uh thing called Mark salms that I experience visually uh when I look at him in the mirror that's the dual aspect monus says these are not two different things not two different uh ontological entities

there's one underlying ontological entity uh and that's called Mark salms so then the question is which is an abstraction then then the question becomes well what stuff is Mark SS made of and I'm saying Mark SS is made of a functional mechanism uh and which the one I was describing earlier uh is a self-organizing therefore in biological terms a homeostatic system uh homeostasis is a mechanism it's an abstraction there's there's there's we infer the underlying mechanisms uh from the the The observed data uh when it comes to the the apparatus of the Mind the thing that we are talking about here which is also the apparatus of the brain it's that apparatus that we're talking about it's the it's the the functional principles governing this part of nature that we infer from the two observables so when it comes to um for example memory you know you can have a memory that's the experience of a memory or you can see optogenetically the neurons that are uh activated uh whilst you are having that same memory these are two perspectives upon one thing which is an abstraction called the memory system or the function of memory so uh I think that that is the actual thing that we're studying in cognitive Neuroscience um although I don't like to call it cognitive Neuroscience because that seems to exclude affec of Neuroscience so the thing the thing that we're studying in what we're let's call it neuros psychology which is a slightly

less prejudiced word um in this in this
mind and brain science I think that the
the um the the ultimate object of study
of neuro psychology is things like the
memory system uh the perceptual system U
the executive system and so on we're
trying to discern the laws uh the
underlying monistic laws uh governing
this part of nature that we can observe
from these two different uh perspectives
both both the objective and the
subjective one now um sorry if I if I'm
getting quite abstract no no no to come
back very useful to come back to the
essential Point um if that is the
framework uh the metaphysical framework
within which I'm working and we all have
to when it comes to the Mind Body
relation uh we have to adopt a
metaphysical uh framework and and it's
good that we are clear about which one
we're adopting that's the one I'm
adopting within that framework it
doesn't make any sense to say how did
the physiological phenomenon cause the
psychological phenomenon they are those
two phenomena are dual manifestations of
a single underlying cause that
underlying cause is the mechanism that I
spoke of earlier which explains them
both um the free energy principle is is
is is such a such a thing it it's a it's
a it's a a principle which governs uh
both how we feel and what brain arousal
mechanisms uh we observe physiologically
in other words what you see with your
eyes and what you feel with your
feelings uh are are both manifestations

of that single underlying mechanism so to say how does the brain process cause uh the feeling uh is to ask the wrong question uh it's to ask a question that assumes a metaphysical relationship which I think uh is is is the wrong assumption it's like saying if I look um with my eyes I see a flash of lightning um if I listen with my ears uh I hear a clap of Thunder uh and I ask the question how does that visual thing uh the flash of lightning cause that auditory thing the clap of Thunder it's to misunderstand that they're actually both caused uh by a a Geo electrical event uh which we describe abstractly using the the the laws of physics uh using the language of physics or or or or or um ultimate forces you know that that that kind of language it's not something that you observe it's the mechanism that you the thing that you infer that underlies it so um when you're saying you're wanting to know how I'm saying I have answered you how when I describ to you what the mechanism is uh and I very important importantly I need to make this clear the mechanism I describ to you the mechanism of feeling that is um which is which which you can observe from two observational perspectives in other words you can observe brain arousal processes uh and you can observe subjectively feelings uh that these are both manifestations of a single mechanism that's the one that I described to you and this is the important top I said very importantly it

also explains why this mean ISM has two
um perceptual surfaces why why it has
two perceptual perspectives it explains
why such a mechanism has a subjective
perspective because not everything
everything geometrically has a has a
subjective perspective in other words
you can say there is there is such a
thing uh as uh the perspective of being
a cup or being a table but that doesn't
mean that there is something it is like
to be a cup or to be a table so I'm
saying the mechanism I've described um
which is which is a a mechanism which
does not uh exist does not explain the
behavior of Cups and tables um is why
there is nothing it is like to be a cup
or a table so the the the remember that
the the underlying manistic
ontological entity that we are studying
is that mechanism which which which we
can study both objectively and
subjectively and I've described why this
particular mechanism has a subjective
aspect now to come to what I understood
to be the third part of or
third aspect of your question um which
is a sort of an evolutionary question
you're saying uh why did this happen
What functionality is added uh why what
is gained by this that that that
couldn't go on in the dark I've already
explained why I believe that mechanism
can't go on in the dark now I want to
explain what what it adds um it is in
fact I have in a way already explained
it but let me articulate it in a
slightly different way um a organism

let's say an organism because we're talking about living things here but there's another question as to whether they necessarily have to be living things but we could say a system but let's just say a biological system an organism um which has needs uh in other words it has its own um biologically viable States and it has to remain within those States in other words it's a an organism which functions homeostatically um it it it needs predictions as to how to satisfy those needs in other words how to how to return to homeostasis when it's deviating from it and we are born with such predictions us living things have innate predictions uh those are called reflexes those precisely are our autonomic functions so um when I um I I have a I have a a constant um need uh to balance my blood gases there's a there's a balance between oxygen and carbon dioxide that's viable uh for a human being and I have a very simple way of meeting that need which is that I breathe automatically um my my uh intercostal muscles expand expand my diaphragm that sucks in oxygen uh and then uh the the the there's the relaxing of that of those muscles and that expels carbon dioxide and there is actually a need detector in your hypothalamus if you want to describe this mechanism anatomically which is constantly monitoring oxygen carbon dioxide oxygen carbon dioxide but there's one

prediction as to how to how to um how to maintain the viable balance and that is this autonomic breathing mechanism now why do we ever become conscious of our need to breathe I know you are now becoming conscious of it because I'm talking about it normally you're not conscious of it um and other words what is added by becoming conscious of it so imagine an an organism equipped with the with the need I've just described uh and with the reflex of prediction the innate prediction the fixed innate instinctual prediction as to how to meet that need that's fine until you place such an organism in an unpredict environment in other words something not predicted by natural selection in other words not the default environment for which this prediction um is an appropriate one the one that you should just breathe at a regular rate in other words for example what happens if you're in a carbon dioxide Filled Room uh you're no longer in the usual uh default environment you're you're in a burning building um and the room is full of carbon dioxide at that point you become conscious of your need need to breathe uh this is the why question that you're asking why uh do you become conscious of your need to breathe it's because now you're in a state of uncertainty as to what to do uh your reflex autonomic uh prediction doesn't cut it uh so you behave stastically you you're in a state of distress uh you you're you you're suffering

um Suffocation alarm um or or or
respiratory distress or air hunger it
goes by various names and why it comes
to Consciousness as a feeling is because
as you randomly go upstairs uh trying to
uh do something other than your
instinctual your normal reflex of action
it feels worse uh and so that was the
wrong decision uh it was a bad choice to
go upstairs and this and and the
organism registers it in the form of
this feeling of Suffocation is getting
worse so you change your mind uh and you
go downstairs instead and there you get
the enormous relief of feeling better
you know being able to being able to
breathe the oxygen so that that feeling
um has guided your voluntary action in
fact has has has made possible what we
call voluntary action uh it's not just
random Behavior because the the your
your existing prediction doesn't cut it
uh what you do is as I said earlier
predictive work in progress in other
words you actually feel your way through
the problem uh and and and come up with
a new prediction which you then lay down
that is the Adaptive advantage that is
what is evolutionarily um added by this
extended form of homeostasis uh in which
you don't just have some sort of Auto
automatic uh regulator of um that that
that keeps you within your viable bonds
you have in addition to those automatic
Regulators you have a further mechanism
uh which kicks in when the automatic
regulator isn't working because you're
not in the environment that it it

expects um and it it enables you to
um to navigate that uh that novel
unpredicted uncertain environment I it's
one of the what I've just said uh one of
the reasons why I I say that that that
Consciousness is felt
uncertainty that's that's a a py way of
describing what it actually is but I
hope that I've answered your question
there by saying what it adds it adds it
adds the the the ability to survive in
unpredicted novel environments prior to
the um evolution of this um mechanism
this extended form of homeostasis this
this subjective form of Home extension
of
homeostasis the what would have happened
um is that if you have a population of
the organisms that um the hypothetical
organisms I'm talking about they all uh
have this stereotyped fixed prediction
as to what to do if they find themselves
in a novel situation because of the
innate variation in the genome some of
them will do the right thing and and
some of them will do the wrong thing in
their in their random sort of thrashing
about uh the few that do the right thing
uh will then pass on uh that uh
adaptation uh to the Next Generation in
other words the Next Generation when
they find themselves in this what was
previously a novel situation that will
not be a novel situation for the Next
Generation the the functionality that
I've described supersedes that that is
the mechanism of natural selection
whereby over Generations um the

predictions are improved by natural selection uh the the the functionality I've described enables the organism within its own lifetime uh to improve its predictive model um in in um in radically uncertain circumstances and thereby to survive and reproduce there's a lot there um so I want to start by being philosophically skeptical and therefore annoying and say that I still think that this is rest on an axiom that for example Consciousness is there for unpredictability because I still can't see why in principle natural selection couldn't have endowed us with just a very good monitor of what to do in uncertain circumstances so I for I can't imagine why a zombie a philosophical zombie wouldn't be able to recognize as I go up the stairs there's more smoke it's getting worse oh let me go downstairs right and this is just you can just keep fleshing out and having recursive iterations on the mechanism before it needs to come on and become online that is and so the way that I thought about it just to sort of put my cards on the table because I've been acting cryptic is I wonder whether um I wonder whether Consciousness acts as a coar grained representation that allows for Speed and manageable data compress and and and in and genders manageable data compression such that we cannot we can't in principle attend to every single data um sources out there in the world but what

Consciousness allows us to do is to compress the data that is in the world into something one that's prioritized and two that can be acted upon in a fast manner whether this is in predictable or unpredictable circumstances and often it will need to be done in unpredictable circumstances but the pure notion that Consciousness is unpredictable is is there for unpredictable circumstances I don't think can skirt the problem or the contention

that hypothetic we can never run the counterfactual but there is a there is a hypothetical counterfactual where all of this is being done offline you just have a very adaptive sensor in the hypothalamus let's say who doesn't who who doesn't need an online self model to recognize any signal in some phenomenal way so that's that again I don't feel we to go down this Rabbit Hole um well well because you have articulated your view I will just say something very briefly about it which is that that is in in in essence and I'm I'm sorry to be so reductionist about you the subtleties in what you've just said but it is in essence a attentional mechanism that you've described it is and uh I'm saying that's fine I spoke earlier about prioritization so that's that's built into what I'm saying uh but uh it's not just any old attentional mechanism it's attention to my homeostatic state in other words to my deviation from my viable bounds it's not attention to anything in the world in other words

just unpredictability I'm saying no it's not just unpredictability it's unpredictability as to how to satisfy this need in other words it's a certain kind of unpredictability for a certain kind of system I think that that is the ground zero of feeling that is how it comes into existence so it is by the mechanism you describe but only applied to a particular part of Nature and not not to everything in the universe that's a that's a very useful ad I've actually just written this paper uh with the help of Carl and L Costa about sort of eliding the endogenous exogenous attention distinction within active inference by saying that attention is basically as as we've been speaking about a priority filter but always for the sense making all organism um and and and in active inference you can say well and then you flesh that out and say well the deepest most precise preferences are encoded at the deepest level of the generative hierarchy hence why I don't care where my phone is when I'm suffocating for example because the preference I have my phone is far less precise and far higher than um that is exactly how I see it yep yeah okay so so I wanted to ask you said something about so we have dual aspect monism that we can describe to phenomena we can take two perspectives on a phenomena like memory for example we can think of the hippocampus and we can also think of what it is like to have a memory and you said that what the real Target of

cognitive Neuroscience or neuros
psychology should be is the single
mechanism that underlies both of those
perspectives and you preferred the
active inference or the free energy
principle might be one of those
mechanisms I wonder whether this was
relating to what people talk about in
ter in Act of inference as a dual
information geometry and what that is is
for the audience and I spoke to Van Visa
about this actually yesterday so the
podcast should be out before this one so
people can go there but the idea is that
we have an intrinsic geometry which
looks at how the actual internal states
of the markov blanket evolve over time
and then we have the E extrinsic
geometry which is what how they are uh
manifesting variational beliefs about
the external States and one is kind one
you can kind of view as if you wanted to
take a metaphysical stance or a
phenomenal stance one is the view from
within and one is the view from without
I wonder whether that Jewel aspect fits
quite nicely onto your jeel aspect
monism or whether you thought about that
yes um what you're saying intuitively
appeals to me
uh when you say I thought about it now
for precisely 10 seconds and it
intuitively appeals to me there's a
paper called um movian monism which is
by Van visa and car so that that's where
this is yes um I I just wonder you know
I I spoke to Van about this yesterday
and I said what were the metaphysical

consequences before we move on May I just say there's something about uh the there's something about because now I'm remembering that paper um and and I'm remembering what I thought when I read it I think that there is something special about um precision optimization in relation to what you're talking about um in that the Precision um assigned to uh the incoming error signal um in relation to the outgoing prediction um is assessed by the system uh to those to those um to that message passing as it were so I think that um there's there's a special place um for precision optimization uh in in in the story that you just described there yeah absolutely yeah I mean in terms of the paper that I just wrote that's exactly it in terms of a more like in terms of a technical Bas graft pomdp schema what we're saying here is sort of higher order preferences govern Precision weighting over What's called the likelihood Matrix the a matrix and that's kind of been you know Felman and Frist and then later papers have proposed that that really is what attention is it's optimizing that Precision waiting yes we not yeah no go only to add that I think a really critical point that I really wanted to stress in that paper is that that's not just Precision per se so Precision I take to be a very strict mathematical construct in uh inverse entropy or inverse covariance whatever you want to say but not all signals which are given a gain you know extra gain synaptic gain

or given attention are only those which are precise I.E which have a clear cause rather we we govern our attention is governed fundamentally by our goals even if that the object of attention isn't necessarily precise right it can be very ambiguous but it's very important I pay attention to whether my garden hose is a garden hose or a snake and that will also be affected for whether I'm in London or whether I'm in Sydney so just to add that I think people it's really important to distinguish attention precision and precision waiting as free individual constructs yes that makes good sense to me um so I uh I I'll put it in the in in more sort of everyday terms that that um uh it has to do because remember I was saying it's a particular type of system and that's another way of saying what you're saying that um if things with respect to the Precision over my policies if things are turning out as expected that's good if uncertainty prevails that's bad and uh so you know this is why it is good to have confidence in your in your policies because it will result in good feelings is absolutely go ahead no but I gathered that you wanted to move on to to another topic well yeah I mean actually just in reference I I like making sure that people know I kind of feel like I should let people know where they can go to read all this stuff so just in terms of the Precision over your policies this is known as the gamma parameter in active inference and if

people want to read about that Casper
hesp 21 2021 paper is a really good
place to start um where he talks about
veilance in terms of this gamma
Precision okay yes I so I um I watched
your two conversations with Lisa Felman
Barrett and greatly enjoyed them uh and
and yeah I thought they were really
fascinating and the Very novel way of
doing scientific discussion because
normally it's very angry and on Twitter
um I thought a really interesting thing
that came up although it was only really
briefly touched upon was kind of the
distinction between interception and
extra
reception and I I found actually I've
never really thought that it could
potentially be quite a blurry line but I
feel like Lisa was Keen to say that some
of the things that you were proposing
were fundamentally extra receptive I a
pin prick or something like that could
also be construed as inter receptive so
if we say okay let's take the pin prick
example and let's talk about aect so
you're not I if I'm say if I'm correct
you wouldn't believe that all affect is
fundamentally interceptive is that a
fair play fair thing to say oh yes I
absolutely disagree with that view but
slightly comically in relation to the
discussion with Lisa felin Barrett that
you're referring to that was the
starting point of that discussion was
that I said I disagree with her and Neil
Seth uh in their in their associating
affect with interception for them

interception is affect and interception
is this other form of perception
remember to them affect is just a modality
of perception it's an interceptive
modality of perception and I was saying
no I don't agree with that at all I
don't think that it is and then I used
the example of pain and I said what
could be more extra receptive you know
than this thing is from the outside
being stuck into my finger um or a
fright you know there's something
happens out there in the world and that
thing out there in the world startles me
so it's

exteroceptive and so I was saying affect
uh clearly is not extra receptive then
she got into these gymnastics that
you're referring to of saying well you
know if you do this then a pinprick is
interceptive and I say well fine you
know if you do this then that's fine but
to me the crucial thing is that that
it's neither the

extracephalic the feeling uh is the
homeostatic deviation homeostats are not
they are not interceptive or EXT
receptive they both they are registering
a a configuration um which it uh which
it infers this uh this is bad for me or
this is good for me in relation to the
particular category of need that we're
talking about so so I don't think you
and Lisa would necessarily disagree on
that if I listen to that conversation
correctly because I think she was keen
and I guess it's a fundamental part of
her theory that there isn't there isn't

necessarily meaning in the data signal
itself and it's only in its uh collision
with the prediction that we get in
basian terms a posterior which relates
to our actual fundamental prior and that
Divergence is what you might call a so I
you wouldn't disagree on
that can can I just make clear there
because you know these disagreements
with Lisa I think are very important uh
for the future of our field by which I
mean by I mean affective science that
that's the field I'm talking about
that's that what I share with Lisa is an
interest in the in the fundamental
nature of affect or what she calls
emotion because as you said at the
beginning affect and emotion and and and
so on these words are used differently
by different people for her a does not
have quality it has veillance and
arousal but it is not
distinguish in in its
intrinsic um so um the you're quite
right in everything you just said there
that she and I don't differ in the
things that you just said there but for
me the crucial point and you can see in
relation to everything that I've said to
you already why it's crucial uh is that
I think that the starting point is that
those homeostats are innate you know
that we are born with categorically
distinct
homeostats um and the I won't repeat
everything I said earlier but it is from
those homeostats uh that the uh
ultimately that the that that the

quality of feeling arises when I was
saying thirst feels different from
hunger and so on and remember I'm saying
these are not only inceptive the Fright
feels different from Pain these are
extracephalic side it's
extracephalic

from this category of need and if you
prioritize that deviation in other words
this is the one that I'm going to
now I'm going to my un this is the most
Salient category of need uh this is the
one where my I'm going to palpate my
uncertainty as I act in the world uh
this is in other words the one that I'm
going to that is now going to govern my
voluntary behavior that the quality of
the feeling uh that governs uh that
action cycle um is going to arise from
the category uh of need that we're
talking about in other words from the
homeostat

her view is that there are no innate
categories of emotion and I've just
explained why I don't agree with that
that I think is the fundamental
difference between her and me and that's
what I felt was that's what I felt was
the fundamental difference and just so I
can get clarity for myself is your view
that there is a strict mapping at least
numerically between the number of
homeostats and the number of basic
categories so is there a homeostat for well
these are the ANP and basic emotion you
know is there a homeostat for uh seeking
is there a homeostat for Rage or lust or
whatever I do believe that there's a a

homeostat for each one of those what he calls basic emotions um but I need to make clear that those what he calls basic emotions um are not the uh sum total of the basic homeostatic need mechanisms uh that are at work in the in the human or or mamalian or for that matter vertebrate organism um the the the um many of the things that we spoke about at the beginning of our conversation things like hunger and thirst and um and um um respiratory feelings and so on these uh are not emotions these are at least they don't feature in Pip's taxonomy of the emotions because for him emotions are not these things he calls them bodily affects so there are many U homeostats not seven there are many many many many many homeostats in the human organism and it's a very telling fact that not all of them give rise to feelings like blood pressure regulation for example there's no feeling of you know how well or badly you're doing in terms of um in terms of um regulating your blood pressure um and and and I'm just using one example there are many such examples there there ones which are transitional which are Norm nor run autonomically and sometimes we feel them and I just gave an example of that when I spoke about respiration that normally it's it's it's handled autonomically but then at under certain circumstances it comes to Consciousness so I'm sorry I'm probably being overly pedantic but the my point is yes I do agree with you that there

for for each need there's a homeostat
but there more than seven of them many
many many more than seven of them and um
only some of them ever come to
Consciousness and only some of them
might be called emotions in my way of
using the word emotion but here's the
further very important Point um the fact
that they are categorical needs doesn't
mean that they aren't amalgams of them
uh it doesn't mean that they also aren't
higher order metacognitive um uh uh um
amalgams and elaborations and so on
for her for for Lisa that's the starting
point the starting point is that there
are these learned categories that we
then um that we then there are what she
calls emotion Concepts which we then
apply to um to what we experience
perceptually and I'm saying yeah of
course there are these learned Concepts
um but they start with uh innate
Concepts those are those are the um
those are the the the homeostats that
I'm talking about how we come to
regulate through Al
estasis and and and and learning um all
of these different mechanisms and and
orchestrate them all with each other and
so on especially us human beings with
all of these higher order abstract
Concepts that we use to do so um it
includes everything that she writes
about I think that what she writes about
describes something that definitely does
happen but I I just I I just question uh
the um the assumption that there are no
innate categories to begin with it just

makes no evolutionary sense and makes no physiological sense it just doesn't map on to what we know about how the brain works um that that's the that's the point I wanted to make yes that was the that was the big sticking point um and there was this kind of back and forth about what evidence is uh which we don't we don't necessarily need to go into but people can watch it and I thought was very interesting yeah well when when if if you do if you dear listener or viewer do watch it I want you to to to notice one thing which I thought was was was especially important um it's that because Lisa's starting point is that this idea that there are particular circuits for particular emotions and so on which which U of course can be paraded and can be caricatured and can be made to look like an idiotic local I lz aist stupid idea but but when she says there are no such systems there are in no innate brain systems for innate emotions and and and other uh categories of organism organismic me when she says that please note in this conversation with me uh I leared that she says the same of um Vision as opposed to sematic sensation as opposed to all faction she says the these two are not innate systems so if that's what she means if if her if her uh um uh way of understanding what we mean by an innate system uh her way of of of using that terminology excludes um the usefulness of speaking about a visual system which we all have and an auditory system that

we all have and so on I think you need to understand that it's in that way that she's questioning whether there are basic emotional systems in the brain uh if if it's such a radical claim that she's that she's making that um we should not speak of any innate systems uh of the kind that we do when we speak of a visual auditory or C sensory and so on system if if if if emotion is being lumped in with that claim then it's not a claim only about emotion it's a whole different way of thinking thinking about what are useful Concepts um and her reasoning for saying there's no innate visual system is that if you're blind your visual system will be um you know will serve different functions not innately visual well of course if you're blind it's going to go in an anomalous is going to organize in an anomalous way but that doesn't mean that it's not useful in biology to speak of the the fact that in the normal state of things there is an an innate visual system but yes I wonder whether I could you know Lisa's not here to State her piece and so people should go and listen and I'd love to have her on um I'm wondering whether I can sort of Steal man and well actually not steal man either yourself or her but also potentially you know offer a middle suggestion which is could we perhaps propose rather than well I wonder whether Lisa would say she might disagree that there are innate Concepts would she disagree that there are innate

processes and by that I mean I can take
a s of active inference biological
active inference stance on this that
there are functions and anatomical
systems in the body and brain which have
priors to self-organized in a certain
way and this is the work of Mike Levin
and France Cookling and this kind of
work about morphospace and how that
fleshed out and so I wonder whether
because I personally and I think this is
an argument that goes all the way back
to Jerry Fodor and the debates around
Chomsky and innate language I can't
see how for example fear would even get
ever get off the ground with no innate
priming for fear right like I agree fear
then becomes socially
modulated but how would that
categorization like I don't think Lisa
would deny that for you and I as fully
fleshed conscious propositionally using
L language users we can think about fear
and have it as a concrete object but how
that ever gets off the ground in terms
of what makes something fear what make
something first without some uh
something guiding that process whether
it's just a singular concept that acts
as a template which things get fit
fitted into or
process uh like let's say an innate
inductive in process something like that
as way that someone like Fodor might
talk about it and arguably where Chomsky
has ended up in in merge and things like
this uh maybe that would give us the
inductive background from which we can

slot these emotions into different categories and in many ways I mean this is this is kind of go you know it takes us all the way to something like a connectionist network which is that these connectionist networks are not completely unsupervised in the sense that they do have prior weights um and so yes I do think the radical tabularasa well one I one I guess I'd have to ask her because again I don't want to speak on her behalf but I wonder whether in steel Manning her position she might have space for something like an innate mechanism or innate process yeah well you're right that um our viewers or listeners must go and watch those they they're on um I think they're on

YouTube they are discussions I had on her Channel yes um and um you so you must watch it yourself and and form you know your let and let her listen to her speaking for herself and and form your own opinion but um the topic that you've just mentioned did come up um not very uh um not very it wasn't discussed at Great length but it came up periodically in our two our two uh conversations and uh my understanding what she said is yes she certainly agrees that there are these innate mechanisms uh and they are instinctual behaviors or reflexive action U not necessarily only external behaviors but also internal actions of within the body um but um all that's innate is those stereotyped uh actions um she doesn't

think or or or believe uh that there is
a feeling that goes with them um so my
my given everything that I said earlier
you can see why I think that some of
them have feeling attached to them
because that's precisely what makes you
know I am you know suffocating as
opposed to I am sleepy uh and uh and you
you act accordingly I I think the
feeling uh attaches to the category of
need um because of what feeling does um
you can I believe um um well I don't I I
don't go into too much detail but let's
just say so I believe that the feeling
is attached to the innate category um
she accepts there are innate categories
but doesn't believe that feelings are
attached to them but here's the really
interesting point she then says so we
then learn we make inferences about
these categories so the state I am in
now which I have been in before I
recognize this state this is called fear
um and that's when fear arises fear is
the concept uh the Learned concept that
you apply to the state that you find
yourself in so the first time uh that
this um uh in my view the first time
that this action uh this
instinctual action freezing and fleeing
occurs um the the the the the feeling
that you will have is fear uh her
position seems to be the first time that
it happens nothing will occur mentally
the second time that it happens you will
recognize I've been in this situation
before I call this fear and now you
formed the concept that's her way of

saying it but the crucial thing there is
there's no place for a feeling in that
why do you have a feeling feeling you've
just categorized this state as fear um
and she says yes indeed it's exactly the
same as any other type of perception
it's perceptual inference and um to me
that leaves qualia entirely out you know
it there there's no need so I asked her
the question why then do we feel uh and
she said I don't go there that's not a
scient
question yes well again uh people should
go and watch and I will email Lisa again
I emailed her and and um I think she was
very very busy she's a very busy woman I
will I will get back in contact so I
couldn't so we're running out of times
we only about 10 minutes left but I
wanted to very briefly touch upon dreams
yes um we haven't spoken we haven't done
anything to do with the kind of earlier
work of yours as I said you know you're
very famous for discovering the forbrain
mechanisms underlying dreams in I think
the late 90s and you wrote in the well
it's I think it's the 2000 paper that
what you're what you're really drawing
into question there is this Association
of REM and dreaming so REM rapid eye
movement sleep and dreaming were natur
well were historically tied together and
you show in that paper that really
nicely there's this sort of double
dissociation that you can have dreams
without REM and REM without dreams and
you said at the very end of that paper
the biological function of dreaming

remains unknown this is at least partly
attributable to the fact that
the function of dreaming and the equally
unknown function of REM sleep have been
conflated for more than 40 Years of
research future studies of these
functions should be uncoupled from one
another I was curious one have they been
uncoupled from one another and two has
that given us any perview or insight
into the function of
dreaming yes um so the double dissoc
thank you for raising this it's that was
my um I mean I spent many uh happy years
uh doing a dream research um and uh I
was very fortunate in the discoveries
that I made uh in in that area which
like most um rarely significant
discoveries happen purely by luck so it
wasn't something I was looking for so I
just want to make sure that that people
understand the double dissociation
you're talking about it's not only that
you can dream outside of REM and you can
be in the REM State without dreaming
that is a is an interesting fact but
it's a that's not a that's I think was
always recognized that the correlation
between dreaming uh and and REM sleep is
a probabilistic one um you are much more
likely to be dreaming in rem than
outside of REM um but um they they the
the thing that I found which was much
more interesting I thought is that
lesions
which destroy the part of the brain that
generates REM which is which is a
colonic system in the mesop pontan mum

that people who who who have lost the capacity to have the REM State because that part of the brain is lesioned uh they continue to dream so with no REM they continue to dream and conversely there are patients um who lose the capacity to dream uh due to a brain lesion elsewhere uh they continue to have REM sleep that's the double dissociation we're talking about standard as well yes indeed and that lesion the one that leads to a loss of dreaming actually there are two lesion sites but the interesting one uh is in the mesocortical mesolimbic dopamine circuitry so it's a different circuit from the the one the the one that generates REM sleep it's a different neurotransmitter from the one that modulates REM sleep so so dreaming uh as opposed to REM sleep they correlate with each other but they're not the same thing um and the the there's an interesting question as to why they should correlate with each other because they do very highly correlate with each other but you can totally dissociate them from each other in the way that I've just described now um I I said which is the quotation you just read that we don't know what the function of dreaming is partly because we conflated it with REM sleep so the research that we were doing for decades um into what is dreaming for the the the causal research that we did the the variable that we manipulated was REM sleep so you know it's not surprising that we never

got to the answer because we were manipulating something that correlates with dreaming but isn't actually the mechanism of dreaming so I said at that point in the paper that you're quoting that now that we can see that they're two separate mechanisms now we can research the mechanism and function of Dreaming by looking at the at at that mechanism not this one um and so uh that was bound to lead to progress um so the the a very interesting you said so has progress been made a lot of progress has been made and in fact um when I said I spent many happy years in a sleep lab I in fact still am doing some research in that area and so there's one study that's still ongoing uh that um that I've got only the I've only completed the pilot study for it but we're now doing the main study but the findings were so dramatic uh in the in the pilot study that I feel able to say a thing or two about them um the but it's not really in the active inference framework so I don't don't worry no no no no no go go go please can be linked and you know Carl and and Hobson uh um Carl Friston and Alan Hobson wrote some very interesting papers within the active inference framework about about dreaming um but the the first thing to say is that uh the this mesocortical mesolimbic dopamine system which clearly is the the driving mechanism the causal mechanism of dreaming and there's all sorts of every method we've thrown at it you know

single cell

recordings uh microdialysis um

pharmacological probes lesion studies

positron emission tomography everything

everything points to that system that

system that system that's the one that

causes the dream state um the the the

really interesting thing about it um is

that it is it's been called uh by Pank

who you men mentioned earlier the

seeking system you know the the in al

behavior that attaches to this system is

foraging exploratory interest and

curiosity U it's it it it it drives the

most sort of intense exploratory

foraging behaviors that an animal is

capable of and when you look at the

dreaming brain for example bosron

emission tomography it's really quite

astounding to see this system is lit up

like a Christmas tree but you're asleep

you know it's like what the hell how can

how can this how can this creature be

asleep or this person be asleep if if

you weren't told that they were asleep

and you just looked at that at that um

at that brain image you would say this

person's um you know Manic and high on

cocaine or something you know so that's

the first thing that's that I think is

really interesting uh is that uh the the

the mechanism that drives the dream uh

is a is an intensely motivated active

State um and so the the the the the

first step toward a an understanding of

why do we dream uh has to be you have to

start there you know why do we have

these intense activated motivated States

while we're sleeping and secondly why don't we wake up uh when we have them um so um one way of thinking uh is that the dream happens instead of real Behavior you have you have a virtual reality in which you are um in which you are imagining or deluding yourself uh that you are acting on your motivated States why these motivated states have to happen while you're asleep is another question but the fact that they do happen is is is obvious physiologically demonstrably true so uh the the this state is acted on in the virtual hallucinatory reality of the dream rather than in real reality because um to act on them in in reality would be incompatible with the state of sleep so that is a that is I think a a very simple and reasonable uh hypothesis as to why do we dream we dream so that we can stay asleep during this highly motivated state of mind that occurs State of Mind and Body the if if it weren't for the fact that we're paralyzed we would be running around like lunatics so you you don't run around like a lunatic you because you're running around in your dream you you so how do you test that hypothesis I said earlier that there are two lesion sites which lead to sensation of dreaming uh and we've spoken only about the one the other one is damage to the visual uh Visos spatial cortices uh at the back of the brain uh if you can't generate the me the mental image the heteromodal and predominantly visual but not only visual

imagery um that is the dream then how
can you dream so
they're theoretically less interesting
those patients they stop dreaming due to
occipitoparietal
Strokes um but if the prediction that
I've just made the hypothesis I've just
formulated as to what the function of
dreaming is if it were correct then
these patients who can't generate the
hallucinated reality because they
because of those those lesions they
should wake up they should have very bad
sleep when when this arousal State
occurs it should wake them up so that's
the prediction that I tested in my pilot
study and let me tell you I have never
seen worse polysomnograph than in these
patient wow yeah so I think we're on to
something there that what what dreams do
is they enable you to stay asleep by
by letting of steam letting of
motivational steam in a in an
hallucinatory World rather than the real
world I'm not saying that's the function
but that's a function that's very cool
that's very cool people have to keep
their eyes out for that for those sets
of papers mark this was awesome we
covered so much ground uh I'm I'm really
really very grateful indeed I always
like to ask people just to say at the
very end where people can find them if
they have you know if people have
questions or people want to read your
work how can people reach out to you or
where should they find your work so uh
to contact me you can contact me by

email um it's just my name Mark somms at
UCT University of Cape Town doac
academic. Za which is South Africa so
mark. somms u.c. Za I am also for my
sins on what used to be called Twitter
but it's now called something else that
I don't want to say um so there there
I'm markor SS at Mark
and we'll make sure to put your Google
Scholar and website and all this stuff
on in the video description again mark
thank you so so very much I enjoyed it
thank you all right